Single Oral Practice 2

Module 5 - Single Oral Administration · Daily practice sheet · No date

Name _____ Date _____

A 10-mg single dose of a drug is given orally to a patient. The drug is about 80% absorbed and the equation that fits the observed data is $C_p = 70$ ng/mL $(e^{-0.1t} - e^{-0.3t})$. [Rate constants are in hr^{-1}]. Determine the following:

Part 1 Volume of distribution

Part 2 Total body clearance

Part 3 Half-life of elimination

Part 4 Half-life of absorption

Part 5 Maximum plasma concentration

Part 6 Maximum plasma concentration if the dose were increased to 20 mg

Part 7 Half-life of elimination if the clearance were reduced to 8.57 L/hr, assuming no change in apparent volume of distribution

Part 8 Maximum plasma concentration for a 10 mg dose if the clearance were reduced to 8.57 L/hr, assuming no change in apparent volume of distribution

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Part 1 Her key

Answer: 171.43 L

WHAT THE STEM GIVES

D_0	10-mg	the dose in the numerator
F	80% absorbed	F = 0.8 in the numerator
A	70 ng/mL	the coefficient, 0.07 mg/L, in the denominator
k	0.1	the smaller exponent, the elimination rate constant, hr^{-1}
k_a	0.3	the larger exponent, the absorption rate constant, hr^{-1}

THE WORKING, ONE LINE AT A TIME

UNIT

$$70\ \text{ng/mL} \times \frac{1000\ \text{mL}}{\text{L}} \times \frac{1\ \text{mg}}{10^6\ \text{ng}} = 0.07\ \text{mg/L}$$

The coefficient is converted from ng/mL to mg/L so that mg ÷ (mg/L) leaves litres for the volume of distribution.

SETUP

$$V_D = \frac{F\,k_a\,D_0}{A(k_a - k)},\ k_a = 0.3,\ k = 0.1$$

In the fitted equation the larger exponent belongs to absorption and the smaller to elimination: $k_a = 0.3$, $k = 0.1\ hr^{-1}$.

ALGEBRA

$$V_D = \frac{(0.8)(10\ \text{mg})(0.3\ hr^{-1})}{(0.07\ \text{mg/L})(0.3 - 0.1)\ hr^{-1}} = 171.43\ \text{L}$$

$(0.8)(10\ \text{mg})(0.3\ hr^{-1}) = 2.4\ \text{mg}\cdot hr^{-1}$ on top; $(0.07\ \text{mg/L})(0.2\ hr^{-1}) = 0.014\ (\text{mg/L})\cdot hr^{-1}$ below. The reciprocal hours cancel, and mg over mg/L leaves litres: $\frac{2.4}{0.014} = 171.43$ L, the volume of distribution V_D that dilutes the absorbed dose to the fitted coefficient.

> **DOES THE NUMBER MAKE SENSE?**
>
> The coefficient 0.07 mg/L is roughly the concentration the 8 mg absorbed would give in a volume of about 100 L, so a V_D in the low hundreds of litres fits: 171.43 L. A value near 0.1 L means ng/mL was not converted to mg/L.

Part 2 Her key

Answer: 17.143 L/hr

WHAT THE STEM GIVES

D_0	10-mg	in the coefficient that gave $V_D = 171.43$ L
F	80% absorbed	in the coefficient that gave V_D

A	70 ng/mL	gave V_D = 171.43 L in part 1
k	0.1	the smaller exponent; $Cl_T = k \times V_D$
k_a	0.3	*not needed: elimination, not absorption, sets clearance*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = V_D \times k = (171.43\ L)(0.1\ hr^{-1}) = 17.143\ L/hr$$

Total clearance $Cl_T = V_D \times k$: the volume of distribution, 171.43 L from part 1, times the fraction of it cleared per hour. In the fitted equation the smaller exponent, 0.1 hr^{-1}, is k, the elimination rate constant, and the larger, 0.3 hr^{-1}, is k_a; (171.43 L)(0.1 hr^{-1}) = 17.143 L/hr.

> **DOES THE NUMBER MAKE SENSE?**
> k = 0.1 hr^{-1} clears a tenth of the volume each hour, so Cl_T is a tenth of 171.43 L: 17.143 L/hr. Using 0.3 instead of 0.1 would give three times too much.

Part 3 Her key

Answer: 6.93 hr

WHAT THE STEM GIVES

D_0	10-mg	*not needed: the half-life depends on k alone*
F	80% absorbed	*not needed*
A	70 ng/mL	*not needed*
k	0.1	the smaller exponent, hr^{-1}: $t½ = \frac{0.693}{k}$
k_a	0.3	*not needed: the larger exponent is absorption*

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693}{0.1\ hr^{-1}} = 6.93\ hr$$

The first-order half-life is 0.693 divided by the elimination rate constant; reciprocal hours in the denominator give hours.

> **DOES THE NUMBER MAKE SENSE?**
> k = 0.1 hr^{-1} means a tenth of the drug leaves each hour, so half is gone in about 7 hours: 6.93 hr. Using the larger exponent 0.3 would give the absorption half-life instead.

Part 4 Her key

Answer: 2.31 hr

WHAT THE STEM GIVES

D_0	10-mg	*not needed: the half-life depends on k_a alone*
F	80% absorbed	*not needed*

A	70 ng/mL	*not needed*
k	0.1	*not needed: the smaller exponent is elimination*
k_a	0.3	the larger exponent, hr^{-1}: $t½a = \frac{0.693}{k_a}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½a = \frac{0.693}{0.3\ hr^{-1}} = 2.31\ hr$$

The absorption half-life is 0.693 divided by the absorption rate constant, the same first-order relation applied to k_a.

> **DOES THE NUMBER MAKE SENSE?**
> $k_a = 0.3\ hr^{-1}$ is three times $k = 0.1\ hr^{-1}$, so the absorption half-life is a third of the elimination half-life: 2.31 hr. Here absorption is faster, so its half-life is the shorter one.

Part 5 Her key

Answer: 26.94 ng/mL

WHAT THE STEM GIVES

D_0	10-mg	*not needed: already inside the 70 ng/mL coefficient*
F	80% absorbed	*not needed: already inside the coefficient*
A	70 ng/mL	the coefficient, multiplied by the bracket at t_{max}
k	0.1	in t_{max} and in $e^{-k\ t_{max}}$
k_a	0.3	in t_{max} and in $e^{-k_a\ t_{max}}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max} = \frac{\ln(0.3/0.1)}{(0.3 - 0.1)\ hr^{-1}} = 5.49\ hr$$

t_{max}, the time of the peak, is when the absorption and elimination rates are equal: $t_{max} = \frac{\ln(k_a/k)}{k_a - k}$, with $k_a = 0.3\ hr^{-1}$ the absorption rate constant and $k = 0.1\ hr^{-1}$ the elimination rate constant. ln 3 = 1.0986, and $\frac{1.0986}{0.2\ hr^{-1}} = 5.49\ hr$; neither dose nor volume enters.

ALGEBRA

$$C_{max} = 70\ ng/mL\ (e^{-(0.1)(5.49)} - e^{-(0.3)(5.49)}) = 26.94\ ng/mL$$

At t = 5.49 hr the exponents are (0.1)(5.49) = 0.549 and (0.3)(5.49) = 1.647, pure numbers; $e^{-0.549} = 0.5775$ and $e^{-1.647} = 0.1926$, so the bracket is 0.3849 and 70 ng/mL × 0.3849 = 26.94 ng/mL, the peak C_{max} in the fitted equation unit, ng/mL.

> **DOES THE NUMBER MAKE SENSE?**

The bracket is a difference of two numbers between 0 and 1, so C_{max} lies below the 70 ng/mL coefficient; at t_{max} = 5.49 hr the bracket is a little over a third, so a peak near 27 ng/mL fits, and 26.94 does.

Part 6 Her key

Answer: 53.88 ng/mL

WHAT THE STEM GIVES

D_0	10-mg	*not needed: replaced by 20 mg*
D_0 new	20 mg	doubles the coefficient: twice the dose, twice the peak
F	80% absorbed	unchanged, in the coefficient
A	70 ng/mL	the coefficient for 10 mg; doubled for 20 mg
k	0.1	in t_{max} and in $e^{-k\,t_{max}}$; unchanged
k_a	0.3	in t_{max} and in $e^{-k_a\,t_{max}}$; unchanged

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max} = \frac{(0.8)(20\text{ mg})(0.3\text{ hr}^{-1})}{(171.43\text{ L})(0.3-0.1)\text{ hr}^{-1}}\left(e^{-(0.1)(5.49)} - e^{-(0.3)(5.49)}\right) = 0.05388\text{ mg/L}$$

The coefficient $\frac{F\,k_a\,D_0}{V_D(k_a - k)}$ is proportional to the dose D_0, so 20 mg doubles it from 0.07 to 0.14 mg/L; t_{max} holds no dose and stays 5.49 hr. The bracket $e^{-0.549} - e^{-1.647} = 0.5775 - 0.1926 = 0.3849$ is unchanged; (0.14 mg/L)(0.3849) = 0.05388 mg/L.

UNIT

$$0.05388\text{ mg/L} \times \frac{1\text{ L}}{1000\text{ mL}} \times \frac{10^6\text{ ng}}{1\text{ mg}} = 53.88\text{ ng/mL}$$

The result equals 2 × 26.94 ng/mL: with first-order kinetics the peak is proportional to the dose, and t_{max} does not move.

DOES THE NUMBER MAKE SENSE?

Doubling the dose from 10 to 20 mg doubles the coefficient and the peak at the same t_{max}, 5.49 hr; twice the part 5 peak is 53.88 ng/mL, still below twice the 70 ng/mL coefficient.

Her key prints 0.0588 mg/L on the first line and 0.05388 on the next; 0.05388 is the value.

Part 7 Her key

Answer: 13.86 hr

WHAT THE STEM GIVES

D_0	10-mg	in the coefficient that gave V_D = 171.43 L
F	80% absorbed	in the coefficient that gave V_D
A	70 ng/mL	gave V_D = 171.43 L in part 1
k	0.1	*not needed: replaced by the new clearance*

k_a	0.3	*not needed*
Cl new	8.57 L/hr	the denominator of $t½ = \frac{0.693\ V_D}{Cl}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693(171.43\ L)}{8.57\ L/hr} = 13.86\ hr$$

$t½ = \frac{0.693\ V_D}{Cl}$ combines $Cl = k \times V_D$ with $k = \frac{0.693}{t½}$. V_D, the volume of distribution, is unchanged at 171.43 L; (0.693)(171.43 L) = 118.8 L, and $\frac{118.8\ L}{8.57\ L/hr} = 13.86$ hr. Clearance fell from 17.143 to 8.57 L/hr, half, so the half-life doubled from 6.93 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Clearance fell to 8.57 L/hr, half of the part 2 value, with the same 171.43 L, so the half-life is twice the part 3 value: 13.86 hr. A smaller clearance always lengthens the half-life.

Part 8 Her key

Answer: 32.61 ng/mL

WHAT THE STEM GIVES

D_0	10 mg	the dose in the numerator
F	80% absorbed	F = 0.8 in the numerator
A	70 ng/mL	*not needed directly: gave V_D = 171.43 L in part 1; (k_a − k) changed*
k	0.1	*not needed: replaced by $k = \frac{8.57}{171.43} = 0.05\ hr^{-1}$*
k_a	0.3	unchanged; in the coefficient, t_{max} and $e^{-k_a t_{max}}$
Cl new	8.57 L/hr	with V_D gives the new k = 0.05 hr^{-1}

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{8.57\ L/hr}{171.43\ L} = 0.05\ hr^{-1}$$

Clearance equals k times the volume of distribution, so dividing clearance by the volume gives the rate constant in reciprocal hours.

ALGEBRA

$$t_{max} = \frac{\ln(0.3/0.05)}{(0.3 - 0.05)\ hr^{-1}} = 7.17\ hr$$

$\frac{0.3}{0.05} = 6$, a pure ratio of k_a to the new k, and ln 6 = 1.792; dividing by (0.3 − 0.05) hr^{-1} = 0.25 hr^{-1} gives $\frac{1.792}{0.25\ hr^{-1}} = 7.17$ hr. The peak comes later than with the original k = 0.1 hr^{-1} because less drug is eliminated while absorption continues.

ALGEBRA

$$C_{max} = \frac{(0.8)(10\ \text{mg})(0.3\ \text{hr}^{-1})}{(171.43\ \text{L})(0.3 - 0.05)\ \text{hr}^{-1}} (e^{-(0.05)(7.17)} - e^{-(0.3)(7.17)}) = 0.03261\ \text{mg/L}$$

The coefficient contains ($k_a - k$), so when k changes the whole oral equation is used rather than the old fitted coefficient.

UNIT

$$0.03261\ \text{mg/L} = 32.61\ \text{ng/mL}$$

The answer is higher than 26.94 ng/mL: with slower elimination, more drug builds up before the peak, so the peak is higher and later.

DOES THE NUMBER MAKE SENSE?

With k halved to 0.05 hr^{-1} less drug is lost during absorption, so the peak is above the part 5 value and comes later, at 7.17 hr; the coefficient rises too, so a peak in the thirties of ng/mL fits: 32.61.